

A Capacity-Gated Boundary Latch: Conditional Coherence and Completion Contracts

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Abstract—We construct a finite occupancy and completion interface for a proposed boundary latch. The historical UD capacity parameter $7/768$ is encoded as a declared threshold, without inferring material enforcement. A normalized two-mode edge-to-face map is isometric, whereas the full coboundary is not. A completion-conditioned Hamiltonian and data-independent decay preserve arbitrary data spectators. For fixed eligible control labels, an exact rational bound places complete system-channel error below 2^{-16} throughout a nonzero readout window. Negative controls show that saturation alone permits premature capture and data-resolving decay destroys captured entanglement. This does not solve native admission, generate the completion flag, implement local couplings or establish physical feasibility.

Index Terms—Quantum interfaces, dissipative control, dark state, completion flag, simplicial boundary map.

I. SOURCE PARAMETER AND FINITE OCCUPANCY CONTRACT

The archived UD manuscript [1] defines $C = 7/(4 \cdot 192) = 7/768$ and a threshold on local amplitude density/backlog. The Tetra blueprint [2] asserts structural saturation as a routing-isolation mechanism. These sources establish specification intent; their statements are not promoted here to physical transport laws, fault tolerance or measured material performance.

Declare a nine-level occupancy register Q with

$$\hat{B} = \sum_{n=0}^8 \frac{n}{768} |n\rangle\langle n|, \quad P_{\text{sat}} = \sum_{n=7}^8 |n\rangle\langle n|. \quad (1)$$

The boundary convention is explicit: equality saturates. These levels are a chosen normalized encoding, not particles per second or measured electron density. A completion flag F has projector $P_F = |1\rangle\langle 1|$. Eligibility is $P_{\text{sat}} \otimes P_F$; saturation and completion are distinct. Occupancy updates and correct flag generation remain external interface assumptions.

This finite-register formulation supplies linear operator dynamics. Replacing it with an instantaneous threshold on an unknown state's expectation is not assumed equivalent to a fixed Lindblad generator. Coherent superpositions across eligible and ineligible control labels are outside the error certificate below.

II. A RESTRICTED EDGE-TO-FACE ISOMETRY

Use ascending vertex labels and oriented edges/faces of a filled tetrahedron. Let B_1 and B_2 be the integer boundary

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matrices; set $D_0 = B_1^\top$ and $D_1 = B_2^\top$. Exact arithmetic verifies

$$D_1 D_0 = 0, \quad \text{rank } D_1 = 3. \quad (2)$$

Therefore ordinary vertex-to-edge-to-face coboundary composition cannot implement a nonzero transfer. Nor can a four-dimensional face register isometrically encode the unrestricted six-dimensional edge register.

Select the two edge basis vectors $01, 02$ with injection $K : \mathbb{C}^2 \rightarrow \mathbb{C}^6$. Define

$$A = D_1 K, \quad A^\dagger A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad S = A(A^\dagger A)^{-1/2}. \quad (3)$$

The Gram eigenvalues are 1, 3, so the inverse square root exists and $S^\dagger S = I_2$ exactly. This preserves a two-mode logical state and its spectators. It is a standard polar normalization of a selected carrier, not a proof that raw coboundary transport preserves quantum information. The transfer can be extended to a unitary with ancillas, but a native autonomous or two-local realization is not constructed here. Normalization changes the operator and is explicitly charged as a proposed interface.

III. COMPLETION-CONDITIONED OPEN DYNAMICS

Use clock levels m, e, d for boundary arrival, auxiliary excitation and dark capture. The stored output is register B . Set $\hbar = 1$ and

$$H = g(P_{\text{sat}} \otimes P_F) \otimes (|m\rangle\langle e| + |e\rangle\langle m|) \otimes I_B, \quad (4)$$

$$L = \sqrt{\Gamma} I_{QF} \otimes |d\rangle\langle e| \otimes I_B, \quad (5)$$

$$\dot{\rho} = -i[H, \rho] + L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}. \quad (6)$$

Assume product launch $|n, f, m\rangle$ independent of arbitrary BR data/reference state, no initial e population, no reverse jumps, and zero storage Hamiltonian. A dark-sector projector commutes with H and is annihilated by L . The data identity factors imply that the entire evolution acts trivially on BR ; this is stronger than clock-projector commutation alone. The Markovian bath is traced out, not included in the certified system channel.

For $n < 7$ or $f = 0$, no amplitude reaches e from this launch and capture probability is exactly zero. For eligible $n \geq 7, f = 1$, the model reduces to the isolated resonant latch. The jump itself does not check completion; safety relies on the launch contract and gated coherent access to e .

The conditional Hamiltonian has full operator norm g . We choose $g = 1, \Gamma = 2\sqrt{2}$. The jump rate is separately declared; a Hamiltonian norm bound does not bound dissipative resources. The projector-conditioned term has not been compiled into geometrically local two-body interactions.

IV. CONDITIONAL COMPLETE-CHANNEL WINDOW CERTIFICATE

Proposition 1. *For fixed eligible labels and the ideal launch contract, complete half-diamond error to the dark-clock identity-data target is less than 2^{-16} for $|t - 16| \leq 1/1024$.*

Proof. No-jump amplitudes on m, e are

$$a(t) = e^{-t/\sqrt{2}}[\cos(t/\sqrt{2}) + \sin(t/\sqrt{2})], \quad (7)$$

$$b(t) = -i\sqrt{2}e^{-t/\sqrt{2}}\sin(t/\sqrt{2}). \quad (8)$$

Exact symbolic differentiation verifies the no-jump equations and initial values. Survival $q = |a|^2 + |b|^2$ obeys

$$q(t) \leq (2 + \sqrt{2})e^{-\sqrt{2}t} < 4e^{-7t/5}. \quad (9)$$

The system clock density is an unnormalized rank-one live block plus $(1 - q)|d\rangle\langle d|$. Its trace distance to the target is exactly q . Tensoring with the identity data channel preserves this equality for complete half-diamond distance, including arbitrary reference registers.

Set $x = 7(16 - 1/1024)/5$ and $E_{20}(x) = \sum_{k=0}^{20} x^k/k!$. Rational arithmetic gives $E_{20}(x) > 2^{18}$, while $e^x > E_{20}(x)$. Throughout the window,

$$q(t) < 4/E_{20}(x) < 2^{-16}. \quad (10)$$

This is an analytic constructor certificate, not a floating simulation estimate. $\square$

V. CONTROLS, RESOURCE ACCOUNTING AND OPEN GATES

The replay has 35 scoped exact checks and 34 numerical controls. Binary64 simulation tests four occupancy labels, both flag values and four times, checking trace, positivity, capture and unnormalized captured Bell negativity. Numerical survival

at $t = 16$ is approximately $3.29 \cdot 10^{-10}$; the rational window bound supplies the proof.

Two negative controls are decisive. Removing completion gating captures an incomplete launch with probability greater than 0.999 at $t = 16$. Replacing the single data-independent jump with separate logical-basis-resolving jumps yields zero captured Bell negativity. An absorbing clock flag therefore does not, by itself, establish coherent storage.

The test model's dimensions are $9 \cdot 2 \cdot 3 \cdot 2 \cdot 2 = 216$, including one data qubit and one spectator. These are register dimensions, not a count of implemented physical qubits. Bath capacity, reverse thermal processes, state preparation, occupancy updates, completion certification and local compilation remain uncharged physical resources. No thermodynamic work bound, graphene/hBN enforcement, geometric noise screening or quantum advantage is established.

The boundary isometry and latch are separate verified contracts; their autonomous composition is OPEN. The existing six-qubit native joint-admission problem is also OPEN. Next, generate and verify the completion flag within a fixed compiled episode, then audit false-completion, coherent control sectors and finite-temperature retention. This note's threshold encoding and deadline are engineering choices; the analytical bound is not an independent prediction or experimental validation.

REFERENCES

- [1] B. W. Mayes, "Unified Dynamics: Derivation of the Continuous Vierbein, Gauge Symmetries, and Electroweak Stability from a Background-Independent Simplicial Substrate," archived manuscript, May 29, 2026, Sec. III. Historical specification source; physical claims not adopted as verified evidence.
- [2] "UD Tetra Quantum architecture," archived two-page architecture blueprint, retrieved October 6, 2026. Proposal source; hardware validation not supplied.